TEST REPORT

ANSI/CAN/UL 9540A:2019
Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems

Report Number............................. :	NN22GLCI.002
Date of issue .... :	07/08/2022
Total number of pages.................. :	28
Name of Testing Laboratory preparing the Report..................... :	**TÜV Rheinland of North America, Inc.** 1279 Quarry Lane, Suite A, Pleasanton, CA 94566
Applicant's name .... :	**Tesla, Inc.**
Address .... :	3500 Deer Creek Road, Palo Alto, CA 94304
Test specification:	
Standard .... :	ANSI/CAN/UL 9540A:2019
Test procedure.............................. :	Report
Non-standard test method........... :	N/A
Test Report Form No. .... :	N/A
Test Report Form(s) Originator .... :	N/A
Master TRF .... :	Dated 2019-01-17

General disclaimer:

The test results presented in this report relate only to the object tested.
This report shall not be reproduced, except in full, without the written approval of the Issuing Testing Laboratory. The authenticity of this Test Report and its contents can be verified by contacting the CB, responsible for this Test Report.

Other / Scope:

1. Add MP2 XL model number in this report

Test item description...................... :	Battery Energy Storage System
Trade Mark .... :	Tesla
Manufacturer.................................. :	Tesla, Inc, (new # 1210368) 3500 Deer Creek Rd, Palo Alto, CA 94304
Model/Type reference..................... :	1748844-XX-Y, 1848844-XX-Y XX – can be any number from 00 to 99. XX – represents style codes used for different variants of the same part, having no impact on the safety and functionality of the entire product. Y – Can be any upper case letter from A to Z. Y – represents pedigree and is used for tracking changes to parts that have already been released to suppliers or production, having no impact on the safety and functionality of the entire product.
Ratings... :	**<u>Megapack 2 - 1748844-XX-Y</u>** 480Vac, 50/60 Hz, 3P3W, Apparent power rating 4hr: 367.5kVA to 997.5 kVA; 2hr: 735kVA to 1890 kVA Nominal Battery capacity 4hr: 1142.4kWh to 3100.8 kWh; 2hr: 1124.2kWh to 2890.8 kWh Nominal Battery power 4hr: 775.2 kW, 2 hr: 1445.4 kW **<u>Megapack 2 XL- 1848844-XX-Y</u>** 480Vac, 50/60 Hz, 3P3W, Apparent power rating 4 hr –504kVA to 1512kVA; 2 hr – 1008kVA to 2400kVA Nominal Battery capacity 4 hr – 1305.6kWh to 3916.8 kWh; 2 hr – 1284.8kWh to 3854.4kWh Nominal Battery power 4 hr – 979.2kW, 2hr - 1927.2kW

Responsible Testing Laboratory (as applicable), testing procedure and testing location(s):

☒	**Testing Laboratory:**	**TÜV Rheinland of North America, Inc.** 1279 Quarry Lane, Suite A, Pleasanton, CA 94566	
Testing location/ address........................... :			
Tested by (name, function, signature)....... :			
Approved by (name, function, signature).. :			
☐	**Testing procedure: CTF Stage 1/TMP:**		
Testing location/ address........................... :			
Tested by (name, function, signature)....... :			
Approved by (name, function, signature).. :			
☒	**Tesla, Inc.**	**Tesla, Inc.**	
Testing location/ address........................... :		Tesla Battery Test Facility Fernley, Nevada	
Tested by (name + signature)..................... :		Kiran Krishnan Kutty	Kiran

Witnessed by (name, function, signature). :	Himanshu Vaidya	Himanshu
Approved by (name, function, signature).. :	Howard Liu	Liu
☐ **Testing procedure: CTF Stage 3/SMT:**		
☐ **Testing procedure: CTF Stage 4:**		
Testing location/ address........................... :		
Tested by (name, function, signature)....... :		
Witnessed by (name, function, signature). :		
Approved by (name, function, signature).. :		
Supervised by (name, function, signature) :		

List of Attachments (including a total number of pages in each attachment):

N/A

Summary of testing:

Tests performed (name of test and test clause):	**Testing location:**
<u>NN22GLCI.001</u> UL 9540A clause 9 – Unit Level Round 2	Tesla, Inc. Tesla Battery Test Facility Fernley, Nevada

Summary of compliance with National Differences (List of countries addressed): N/A

☐ **The product fulfils the requirements of _________ (insert standard number and edition and delete the text in parenthesis, leave it blank or delete the whole sentence, if not applicable)**

Possible test case verdicts:	
- test case does not apply to the test object ... :	N/A
- test object does meet the requirement .... :	P (Pass)
- test object does not meet the requirement ... :	F (Fail)
Testing:	
Date of receipt of test item .... :	February 22, 2022
Date (s) of performance of tests...................... :	March 09, 2022

General remarks:

"(See Enclosure #)" refers to additional information appended to the report.
"(See appended table)" refers to a table appended to the report.

Throughout this report a ☐ comma / ☒ point is used as the decimal separator.

Name and address of factory (ies) .... :

Copy of marking plate: Use – "Only for use with Tesla Products"

"The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCB' s that own these marks"

General product information and other remarks:

Megapack is bi-directional, supporting charge and discharge. It converts power for storage in rechargeable lithium-ion battery packs (battery modules) and is designed in a modular fashion in order to support a range of AC power.

Each Megapack contains up to 19 battery modules with inverters (24 for Megapack 2 XL), a thermal bay and associated thermal roof components, an AC circuit breaker, and a set of customer interface terminals and internal controls circuit boards. An external auxiliary power supply is not required for Megapack; Megapack pulls auxiliary power for the control power and thermal management from the local AC.

Depending on the system configuration (2-hour or 4-hour), each Megapack can be configured with different quantities of battery modules which, together with the site's grid voltage, determine Megapack's nominal power rating.

Report History:

NN22EDSI.001: UL 9540A clause 9 – Unit Level Round 1 for Megapack 2

NN22GLCI.001: UL 9540A clause 9 – Unit Level Round 2 for Megapack 2

NN22GLCI.002: Add Megapack 2 XL model number

"MP2XL(1848844-XX-Y) uses the same batteries, converters, vents and sparker system used in MP2 (1748844-XX-Y). The enclosure is of similar construction to that of MP2. Key difference in the 2 models is that MP2XL contains 24 AC battery modules instead of 19 in MP2. Based on the limited module propagation observed during MP2 testing (7 cells in runaway) the behavior would be the same with MP2XL. With the increase in volume and sparker count, the deflagration risk is minimized. The testing performed on MP2 is considered harsher with higher gas concentrations, and fundamental engineering analysis for MP2XL shows comparable behavior as worst case"

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict

CONSTRUCTION			--
5	**General**		--
5.1	**Cell**		
5.1.1	The cells associated with the BESS that were tested shall be documented in the test report	CATL Lithium Iron Phosphate Rated Capacity (Ah): 157.2Ah Nominal Voltage: 3.22V Nominal mass: 2991 ± 60 g Dimension: 50.75±0.5mm*166±0.5mm*169.3±0.5mm	
5.1.2	The cell documentation included in the test report shall indicate if the cells associated with the BESS comply with UL 1973	Battery module and cell is certified to UL 1973	P
5.1.3	Refer to 7.6.1 for further details	See 7.6.1	N/A
5.2	**Module**		
5.2.1	The modules associated with the BESS that were tested shall be documented in the test report	Battery module (3 modules in series, 336S1P)	P
5.2.2	The module documentation included in the test report shall indicate if the modules associated with the BESS comply with UL 1973	Battery module is compliant with UL 1973	P
5.2.3	Refer to 8.3 for further details	See 8.3	N/A
5.3	**Battery energy storage system unit**		P
5.3.1	The BESS unit documentation included in the test report shall indicate the units that comply with UL 9540	UL 9540 compliant and certified	P
5.3.2	For BESS units for which UL 9540 compliance cannot be determined,	See above	N/A
5.3.3	If applicable, the details of any fire detection and suppression systems that are an integral part of the BESS shall be noted in the test report	No fire detection and suppression systems used	N/A
5.3.4	Refer to 9.7, 10.4 and 10.7 for further details	See 9.7	N/A
5.4	**Flow Batteries**		N/A
5.4.1	For flow batteries, the report will cover the chemistry, as well as the electrical rating in capacity and nominal voltage of the cell stack	Not flow Batteries	N/A
5.4.2	The flow battery documentation included in the test report shall indicate if the flow battery system complies with UL 1973		N/A
5.4.3	See 7.6.2 for further details		N/A
PERFORMANCE			--
6	**General**		N/A

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Clause	Requirement – Test	Result – Remark	Verdict
6.1	The tests in this standard are extreme abuse conditions conducted on electrochemical energy storage devices that can result in fires		N/A
6.2	At the conclusion of testing, samples shall be discharged in accordance with the manufacturer's specifications		N/A
9	**Unit Level**		
9.1	**Sample and test configuration**		P
9.1.1	The unit level test shall be conducted with BESS units installed as described in the manufacturer's instructions and this section. Test configurations include the following:		P
	a) Indoor floor mounted non-residential use BESS; b) Indoor floor mounted residential use BESS; c) Outdoor ground mounted non-residential use BESS; d) Outdoor ground mounted residential use BESS; e) Indoor wall mounted non-residential use BESS; f) Indoor wall mounted residential use BESS; g) Outdoor wall mounted non-residential use BESS; h) Outdoor wall mounted residential use BESS; and i) Rooftop and open garage non-residential use BESS installations.	Outdoor ground mounted non-residential use BESS	P
9.1.2	The unit level test requires one initiating BESS unit in which an internal fire condition in accordance with the module level test is initiated and target adjacent BESS units representative of an installation	One initiating BESS and three target adjacent BESS	P
	Exception: Testing can be conducted outdoors for outdoor only installations if there are the following controls and environmental conditions in place:	Testing can be conducted outdoors for outdoor only installations See Figure 1	P
	a) Wind screens are utilized with a maximum wind speed maintained at ≤ 12 mph; b) The temperature range is within 10°C to 40°C (50°F to 104°F); c) The humidity is < 90% RH; d) There is sufficient light to observe the testing; e) There is no precipitation during the testing; f) There is control of vegetation and combustibles in the test area to prevent any impact on the testing and to prevent inadvertent fire spread from the test area; and g) There are protection mechanisms in place to prevent inadvertent access by unauthorized persons in the test area and to prevent exposure of persons to any hazards as a result of testing.	This was an outdoor installation test. The ambient temperature was varied between 10°C and 15°C and humidity less than 90% RH, and wind was under 10 mph.	P
9.1.3	Depending upon the configuration and design of the BESS (e.g. the BESS is composed of multiple separate parts within separate enclosures), this testing to determine fire characterization can be done at the battery system level	Testing performed at BESS level	N/A
9.1.4	The initiating BESS unit shall contain components representative of a BESS unit in a complete installation.	Complete unit in the testing	P

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Clause	Requirement – Test	Result – Remark	Verdict

9.1.5	Target BESS units shall include the outer cabinet (if part of the design), racking, module enclosures, and components		P
9.1.6	The initiating BESS unit shall be at the maximum operating state of charge (MOSOC),	100% SOC	P
9.1.7	If a BESS unit includes an integral fire suppression system, there is an option of providing this with the DUT	No integral fire suppression system	N/A
9.1.8	Electronics and software controls such as the battery management system (BMS) in the BESS are not relied upon for this testing.		P
9.2	**Test method – Indoor floor mounted BESS units**	Outdoor ground mounted units. Used the test method described in the Section 9.2 except conflicted with Section 9.3.	-
9.2.1	Samples and test configurations are in accordance with 9.1.	Testing conducted outdoor	N/A
9.2.2	Any access door(s) or panels on the initiating BESS unit and adjacent target BESS units shall be closed,	Doors closed	P
9.2.3	The initiating BESS unit shall be positioned adjacent to two instrumented wall sections	Wall construction per 9.3.3	N/A
9.2.4	Instrumented wall sections shall extend not less than 0.49 m (1.6 ft) horizontally beyond the exterior of the target BESS units.	Wall construction per 9.3.3	N/A
9.2.5	Instrumented wall sections shall be at least 0.61-m (2-ft) taller than the BESS unit height	Wall construction per 9.3.3	N/A
9.2.6	The surface of the instrumented wall sections shall be covered with 16-mm (5/8-in) gypsum wall board and painted flat black	Wall construction per 9.3.3	N/A
9.2.7	The initiating BESS unit shall be centered underneath an appropriately sized smoke collection hood of an oxygen consumption calorimeter	Testing conducted outdoor	N/A
9.2.8	The light transmission in the calorimeter's exhaust duct shall be measured using a white light source and photo detector for the duration of the test	Testing conducted outdoor	N/A
9.2.9	The chemical and convective heat release rates shall be measured for the duration of the test, using the methodologies specified in 8.2.11 and 9.2.12, respectively	Testing conducted outdoor	N/A
9.2.10	With reference to 9.2.9, the heat release rate measurement system shall be calibrated	Testing conducted outdoor	N/A
9.2.11	With reference to 9.2.9, the convective heat release rate shall be measured using thermopile	Testing conducted outdoor	N/A
9.2.12	With reference to 9.2.9, the convective heat release rate shall be calculated using the following equation: $HRR_c = V_e A \frac{353.22}{T_e} \int_{T_0}^{T} C_p dT$	Testing conducted outdoor	N/A
9.2.13	The physical spacing between BESS units (both initiating and target) and adjacent walls shall be representative of the intended installation	Wall construction per 9.3.3	N/A
9.2.14	Separation distances shall be specified by the manufacturer for distance between:		P

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Clause	Requirement – Test	Result – Remark	Verdict
	a) The BESS units and the instrumented wall sections; and b) Adjacent BESS units.	a) 5 ft from Initiator to the left b) 6 inches from ISO knuckle of Initiating unit to Target unit. 4 inches from surface of initiating unit to target unit surface.	P
9.2.15	Wall surface temperature measurements shall be collected for BESS intended for installation in locations with combustible construction.	Wall construction per 9.3.3	N/A
9.2.16	Wall surface temperatures shall be measured in vertical array(s) at 152-mm (6-in) intervals for the full height of the instrumented wall sections using No. 24-gauge or smaller,	Wall construction per 9.3.3	N/A
9.2.17	Thermocouples shall be secured to gypsum surfaces by the use of staples placed over the insulated portion of the wires	Wall construction per 9.3.3	N/A
9.2.18	Heat flux shall be measured with the sensing element of at least two water-cooled Schmidt-Boelter gauges at the surface of each instrumented wall:	Wall construction per 9.3.3	N/A
	a) Both are collinear with the vertical thermocouple array;		N/A
	b) One is positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module; and		N/A
	c) One is positioned at the elevation estimated to receive the greatest heat flux during potential propagation of thermal runaway within the initiating BESS unit.		N/A
9.2.19	Heat flux shall be measured with the sensing element of at least two water-cooled Schmidt-Boelter gauges at the surface of each adjacent target BESS unit that faces the initiating BESS unit:		P
	a) One is positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module within the initiating BESS; and		P
	b) One is positioned at the elevation estimated to receive the greatest surface heat flux due to the thermal runaway of the initiating BESS.		P
9.2.20	For non-residential use BESS, heat flux shall be measured with the sensing element of at least one water-cooled Schmidt-Boelter gauge	Testing conducted outdoor	N/A
9.2.21	No. 24-gauge or smaller, Type-K exposed junction thermocouples shall be installed to measure the temperature of the surface	No. 24-gauge, Type-K used	P
9.2.22	For residential use BESS, the DUT shall be covered with a single layer of cheese cloth	Non-residential	N/A

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Clause	Requirement – Test	Result – Remark	Verdict
9.2.23	An internal fire condition in accordance with the module level test shall be created within a single module in the initiating BESS unit:	See Figure 2 Megapack can consist up to 19 Battery Module assemblies. Each module assembly contains 3 battery Modules each which is a total of 112 cell modules. Two sections of heaters for 2 cells was setup to force thermal runaway.	P
	a) The position of the module shall be selected to present the greatest thermal exposure		P
	b) The setup (i.e. type, quantity and positioning) of equipment for initiating thermal runaway in the module shall be the same as that used to initiate and propagate thermal runaway within the module level test		P
9.2.24	The composition, velocity and temperature of the initiating BESS unit vent gases shall be measured within the calorimeter's exhaust duct	Testing conducted outdoor	N/A
9.2.25	The hydrocarbon content of the vent gas shall be measured using flame ionization detection	Testing conducted outdoor	N/A
9.2.26	The test shall be terminated if:		P
	a) Temperatures measured inside each module within the initiating BESS unit return to ambient temperature;	Applicable	P
	b) The fire propagates to adjacent units or to adjacent walls; or		N/A
	c) A condition hazardous to test staff or the test facility requires mitigation		N/A
9.2.27	For residential use systems, the gas collection data gathered in 9.2 shall be compared to the smallest room installation	Non-residential	N/A
9.3	**Test method – Outdoor ground mounted units**		
9.3.1	Outdoor ground mounted non-residential use BESS being evaluated for installation in close proximity to buildings shall use the test method described in Section 9.2	See 9.2	P
9.3.2	except as noted in 9.3.3 and 9.3.4. Heat flux measurements for the accessible means of egress shall be measured in accordance with 9.2.20.	See 9.2	P
9.3.3	Test samples shall be installed as shown in Figure 9.2 in proximity to an instrumented wall section that is 3.66-m (12-ft) tall with a 0.3-m (1-ft) wide horizontal soffit		P
	Exception: If the manufacturer requires installation against non-flammable material, the test setup may include manufacturer recommended backing material between the unit and plywood wall		N/A
9.3.4	Target BESS shall be installed on each side of the initiating BESS in accordance with the manufacturer's installation specifications	Target unit on the front, side and back of the initiating BESS	P
9.4	**Test Method – Indoor wall mounted units**	Outdoor	N/A

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Clause	Requirement – Test	Result – Remark	Verdict
9.4.1	Testing of indoor wall mounted BESS shall be in accordance with Section 9.2, except as modified in this section. See Figure 9.3.		N/A
9.4.2	The test shall be conducted in a standard NFPA 286 fire test room, 3.66 × 2.44 × 2.44-m (12 × 8 × 8-ft) high, with a 0.76 × 2.13-m (2-1/2 × 7-ft) high opening.		N/A
9.4.3	The initiating BESS unit shall be positioned on the wall opposite of the door opening		N/A
9.4.4	Target BESS shall be installed on the wall on each side of the initiating BESS		N/A
9.4.5	The wall on which the initiating and target BESS units are mounted shall be instrumented in accordance with Section 9.2.		N/A
9.4.6	The gas collection methods shall be in accordance with 9.2		N/A
9.4.7	For residential use BESS, the DUT shall be covered with a single layer of cheese cloth ignition indicator.		N/A
9.5	**Test Method – Outdoor wall mounted units**	Outdoor and ground mounted	N/A
9.5.1	Testing of outdoor wall mounted BESS shall be in accordance with Section 9.2, except as modified in this section. See Figure 9.4.		N/A
9.5.2	Test samples shall be mounted on an instrumented wall section that is 3.66-m (12-ft) tall with a 0.3-m (1-ft) wide horizontal soffit (undersurface of the eave shown in Figure 9.4).		N/A
9.5.3	The initiating BESS unit shall be positioned on the instrumented wall, with its center located 1.22-m (4-ft) above the floor,		N/A
9.5.4	Target BESS shall be installed on the wall on each side of the initiating BESS, at the same height		N/A
9.5.5	The wall on which the initiating and target BESS units are mounted shall be instrumented in accordance with Section 9.2.		N/A
9.5.6	For residential use BESS, the DUT shall be covered with a single layer of cheese cloth		N/A
9.6	**Rooftop and open garage installations**	Outdoor and ground mounted	N/A
9.6.1	Testing of BESS intended for non-residential use rooftop or open garage installations shall be in accordance with 9.2.		N/A
9.6.2	If intended for rooftop and open garage use only installations, the smoke release rate, the convective and chemical heat release rate and content, velocity and temperature of the released vent gases need not be measured		N/A
9.7	**Unit level test report**		
9.7.1	The report on the unit level testing shall identify the type of installation being tested, as follows:		P

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Clause	Requirement – Test	Result – Remark	Verdict
	a) Indoor floor mounted non-residential use BESS; b) Indoor floor mounted residential use BESS; c) Outdoor ground mounted non-residential use BESS; d) Outdoor ground mounted residential use BESS; e) Indoor wall mounted non-residential use BESS f) Indoor wall mounted residential use BESS; g) Outdoor wall mounted non-residential use BESS; h) Outdoor wall mounted residential use BESS; i) Rooftop installed non-residential use BESS; or j) Open garage installed non-residential use BESS.	Outdoor ground mounted non-residential use BESS;	P
9.7.2	With reference to 9.7.1, if testing is intended to represent more than one installation type, this shall be noted in the report	One installation type only	P
9.7.3	The report shall include the following, as applicable:	See table 1	P

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Clause	Requirement – Test	Result – Remark	Verdict
	a) Unit manufacturer name and model number (and whether UL 9540 compliant); b) Number of modules in the initiating BESS unit; c) The construction of the initiating BESS unit per 5.3; d) Fire protection features/detection/suppression systems within unit; e) Module voltage(s) corresponding to the tested SOC; f) The thermal runaway initiation method used; g) Location of the initiating module within the BESS unit; h) Diagram and dimensions of the test setup including mounting location of the initiating and target BESS units, and the locations of walls, ceilings, and soffits; i) Observation of any flaming outside the initiating BESS enclosure and the maximum flame extension; j) Chemical and convective heat release rate versus time data; k) Separation distances from the initiating BESS unit to target walls (e. g. distances A and C in Figure 9.1); l) Separation distances from the initiating BESS unit to target BESS units (e.g. distances D and H in Figure 9.1); m) The maximum wall surface and target BESS temperatures achieved during the test and the location of the measuring thermocouple; n) The maximum ceiling or soffit surface temperatures achieved during the indoor or outdoor wall mounted test and the location of the measuring thermocouple; o) The maximum incident heat flux on target wall surfaces and target BESS units; p) The maximum incident heat flux on target ceiling or soffit surfaces achieved during the indoor or outdoor wall mounted test; q) Gas generation and composition data; r) Peak smoke release rate and total smoke release data; at which activation occurred; t) Observation of flying debris or explosive discharge of gases; u) Observation of re-ignition(s) from thermal runaway events; v) Observation(s) of sparks, electrical arcs, or other electrical events; w) Observations of the damage to: 1) The initiating BESS unit; 2) Target BESS units; 3) Adjacent walls, ceilings, or soffits; and x) Photos and video of the test.		
9.8	**Performance at unit level testing**		

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Clause	Requirement – Test	Result – Remark	Verdict
9.8.1	Installation level testing in Section 10 is not required if the following performance conditions outlined in Table 9.1 are met during the unit level test.	Conditions in table 9.1 met	P
10	**Installation Level**	Unit level testing only	N/A
10.1	General		N/A
10.1.1	The installation level test method assesses the effectiveness of the fire and explosion mitigation methods for the BESS in its intended installation		N/A
	a) Test Method 1 – "Effectiveness of sprinklers" is used		N/A
	b) Test Method 2 – "Effectiveness of fire protection plan" is used		N/A
10.1.2	Installation level testing is not appropriate for units only intended for outdoor use or residential use.	Outdoor only	P
10.2	Sample		N/A
10.2.1	The samples (initiating BESS and target BESS) and their preparation for testing		N/A
10.2.2	A flame indicator consisting of a cable tray with fire rated cables that complies with UL 1685 and representative of the installation per the anufacturer's specifications		N/A
10.3	Test method 1 – Effectiveness of sprinklers		N/A
10.3.1	For BESS units with a height of 2.44 m (8 ft) or less, the test shall be conducted in a 6.10 × 6.10 × 3.05-m (20 × 20 × 10-ft) high test room		N/A
10.3.2	The test room shall be fitted with four sprinklers at 3.05-m (10-ft) spacing in the center		N/A
10.3.3	Walls shall be constructed with 16-mm (5/8-in) gypsum wall board		N/A
10.3.4	The initiating BESS unit shall be positioned at manufacturer specified distances		N/A
10.3.5	Temperature measurements at the ceiling locations directly above the initiating and target BESS unit shall be collected by an array of thermocouples		N/A
10.3.6	Instrumented wall surface temperature measurements shall be collected in a vertical array at 152-mm (6-in) intervals		N/A
10.3.7	Thermocouples for wall surface temperature measurements shall be secured to gypsum surfaces by the use of staples		N/A
10.3.8	Heat flux shall be measured with the sensing element of at least two water-cooled Schmidt-Boelter gauges at the surface of each instrumented wall:		N/A
	a) Both are collinear with the vertical thermocouple array;		N/A
	b) One is positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module; and		N/A
	c) One is positioned at the elevation estimated to receive the greatest heat flux during potential propagation of thermal runaway within the initiating BESS unit.		N/A

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Clause	Requirement – Test	Result – Remark	Verdict
10.3.9	Heat flux shall be measured with at least two sensing water-cooled Schmidt-Boelter gauges at the surface of each adjacent target BESS unit that faces the initiating BESS unit:		N/A
	a) One is positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module within the initiating BESS; and		N/A
	b) One is positioned at the elevation estimated to receive the greatest surface heat flux due to the thermal runaway of the initiating BESS.		N/A
10.3.10	The heat flux shall be measured with the sensing element of at least one water-cooled Schmidt-Boelter gauge		N/A
10.3.11	No. 24-gauge or smaller Type-K exposed junction thermocouples shall be installed		N/A
10.3.12	An internal fire condition in accordance with the module level test shall be created		N/A
	a) The position of the module shall be selected to present the greatest thermal exposure		N/A
	b) The setup (i.e. type, quantity and positioning) of equipment for initiating thermal runaway in the module shall be the same		N/A
10.3.13	The composition of BESS unit vent gases shall be measured		N/A
10.3.14	The test shall be terminated if:		N/A
	a) Temperatures measured inside each module of the initiating BESS return to below the cell vent temperature;		N/A
	b) The fire propagates to adjacent units or to adjacent walls; or		N/A
	c) A condition hazardous to test staff or the test facility requires mitigation.		N/A
10.3.15	The initiating unit shall be under observation for 24 h after conclusion of the installation test		N/A
10.4	**Installation level test report – Test method 1 – Effectiveness of sprinklers**	No sprinkler system	N/A
10.4.1	The report on installation level testing shall include the following:		N/A

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Clause	Requirement – Test	Result – Remark	Verdict
	a) Unit manufacturer name and model number (and whether compliant with UL 9540); b) Number of modules in the initiating BESS unit; c) The construction of the initiating BESS unit per 5.3; d) Module voltage(s) of initiating BESS corresponding to the tested SOC; e) The thermal runaway initiation method used; f) Diagram and dimensions of the test setup including location of the initiating and target BESS units, and the locations of walls and ceilings; g) Location of initiating module within the BESS unit; h) Separation distances from the initiating BESS unit to (e.g. distances A and C in Figure 10.1); i) Separation distances from the initiating BESS unit to target BESS units (e.g. distances D and E in Figure 10.1); j) Distances of the flame indicator (if used) with respect to the BESS (e. g. distances A and B in Figure 10.2); k) Maximum temperature at the ceiling; l) Distance of fire spread within the flame indicator; m) The maximum wall surface and target BESS unit temperatures achieved during the test and the location of the measuring thermocouple; n) The maximum incident heat flux on target wall surfaces and target BESS units; o) Voltages of initiating BESS; p) Total number of sprinklers that operated and length of time the sprinklers operated during the test; q) Gas generation and composition data, if measured; r) Observation of flaming outside of the test room s) Observation of flying debris or explosive discharge of gases; t) Observation of re-ignition(s) from thermal runaway events; u) Observations of the damage to: 1) The initiating BESS unit; 2) Target BESS units; and 3) Adjacent walls; v) Photos and video of the test; w) Fire protection features/detection/suppression systems within unit; and x) Sprinkler K-factor, RTI, manufacturer and model, number of sprinklers and layout		N/A
10.5	**Performance – Test method 1 – Effectiveness of sprinklers**		N/A
10.5.1	For BESS units intended for installation in locations with combustible construction, surface temperature measurements along instrumented wall surfaces shall not exceed a temperature rise of 97°C		N/A
10.5.2	The surface temperature of modules within the BESS units adjacent to the initiating BESS unit shall not exceed the temperature at which thermally initiated cell venting occurs		N/A

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict

10.5.3	The fire spread on the cables in the flame indicator shall not extend horizontally beyond the initiating BESS enclosure dimensions		N/A
10.5.4	There shall be no flaming outside the test room.		N/A
10.5.5	There is no observation of detonation.		N/A
10.5.6	Heat flux in the center of the accessible means of egress shall not exceed 1.3 kW/m2.		N/A
10.5.7	There shall be no observation of re-ignition within the initiating unit after the installation test		N/A
10.5.8	An installation level test that does not meet the applicable performance criteria noted above is considered noncompliant and would need to be revised and retested		N/A
10.6	**Test method 2 – Effectiveness of fire protection plan**	Refer 10.3	N/A
10.6.1	The test method 2 test set-up and test procedures are identical to that in 10.3, except instead of the sprinkler system set up of 10.3.2, the room shall be fitted with the specified fire protection		N/A
10.7	**Installation level test report – Test method 2 – Effectiveness of fire protection plan**	Refer 10.4	N/A
10.7.1	The report on installation level testing shall include the following:		N/A
	a) The report information in 10.4.1 items (a) – (u), and (v) if applicable; b) Fire protection features/detection/suppression systems within installation; and c) Length of time of operation of the clean agent, or other suppression system in addition to any sprinklers used.		N/A
10.8	**Performance – Test method 2 – Effectiveness of fire protection plan**	Refer 10.5	N/A
10.8.1	See 10.5 for performance criteria		N/A

ANNEX A	**Test Concepts And Application Of Test Results To Installations**	INFORMATIVE	
A2.1	**General**		N/A
A2.2	**Cell level testing**	CATL UL9540a certified by UL project no. 4789956341	P
A2.3	**Module level testing**	CATL UL9540a certified by TUV Sud project no. 64.290.21.30597.01	P
A2.4	**Unit level testing**	Refer Table 1 below for results	P

ANNEX B	**Safety Recommendations for Testing**	INFORMATIVE	

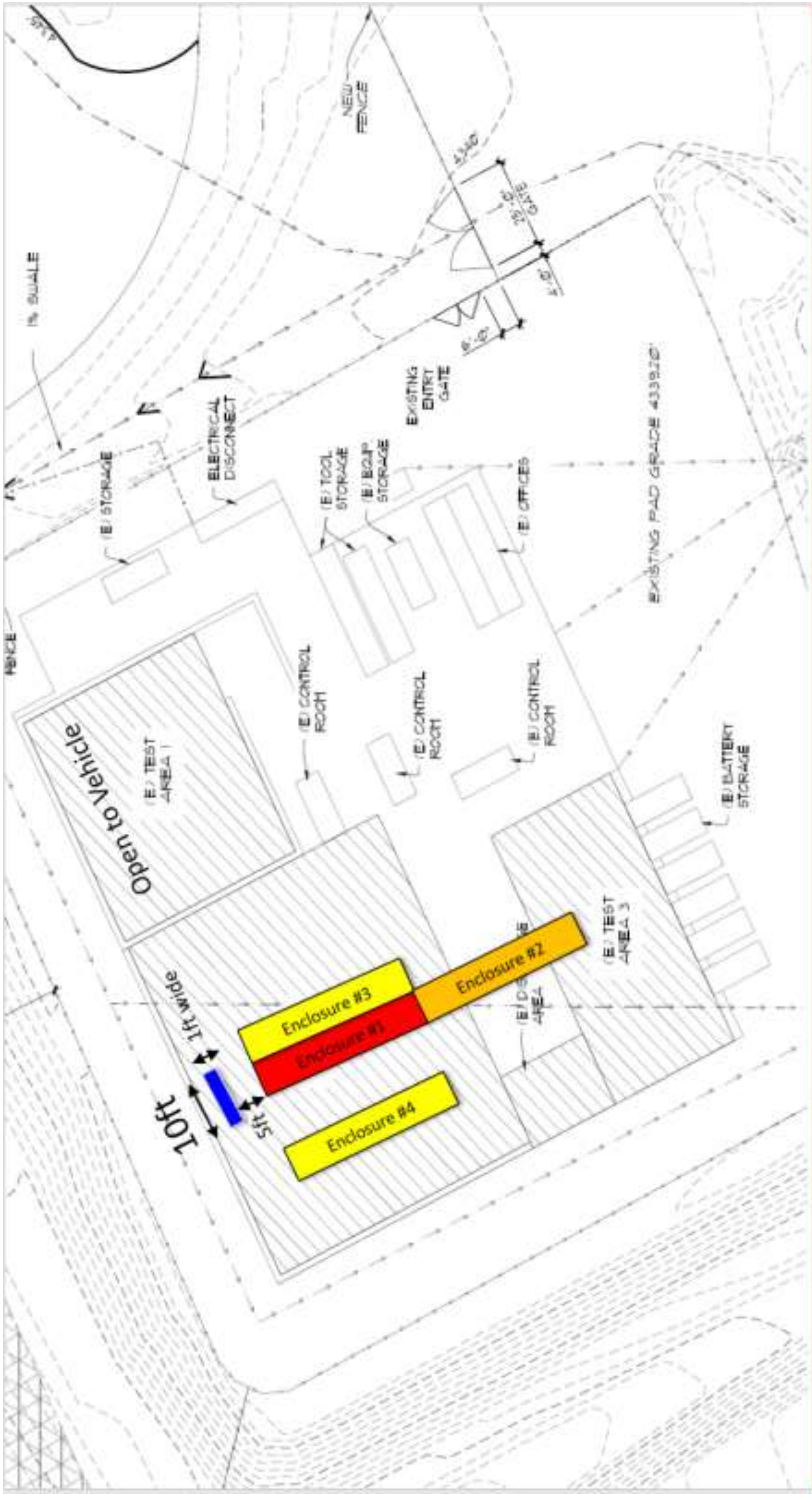

Fig 1. Site Layout

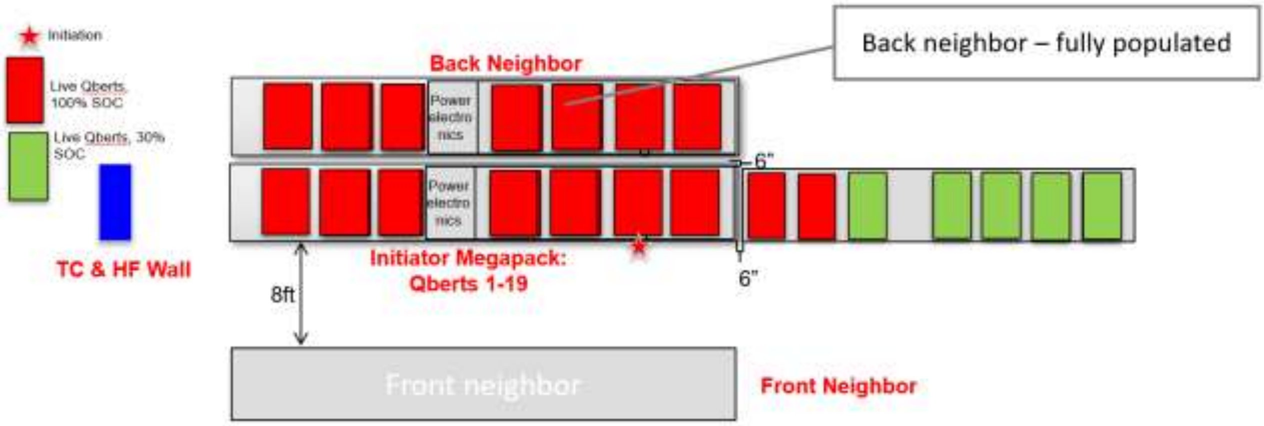

Fig 2. Test area Layout

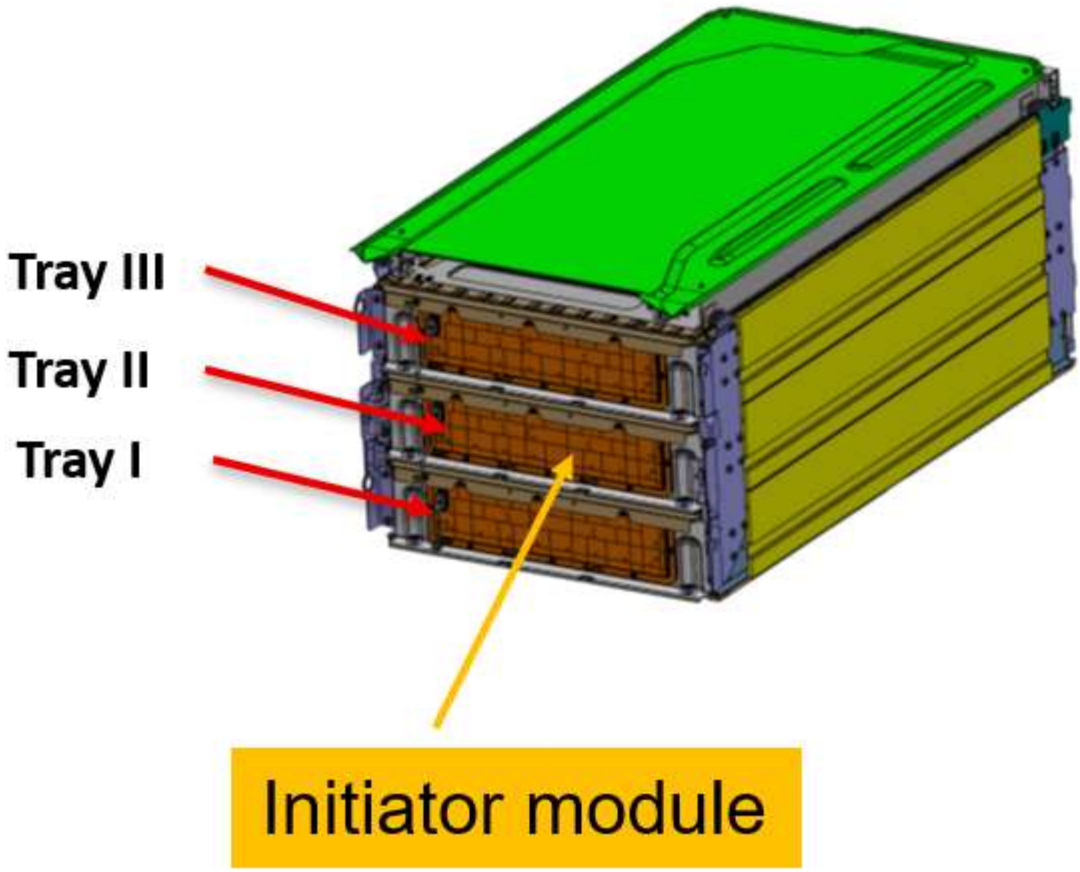

Fig 3. Initiator module location

Fig 4. Initiator AC Battery module location

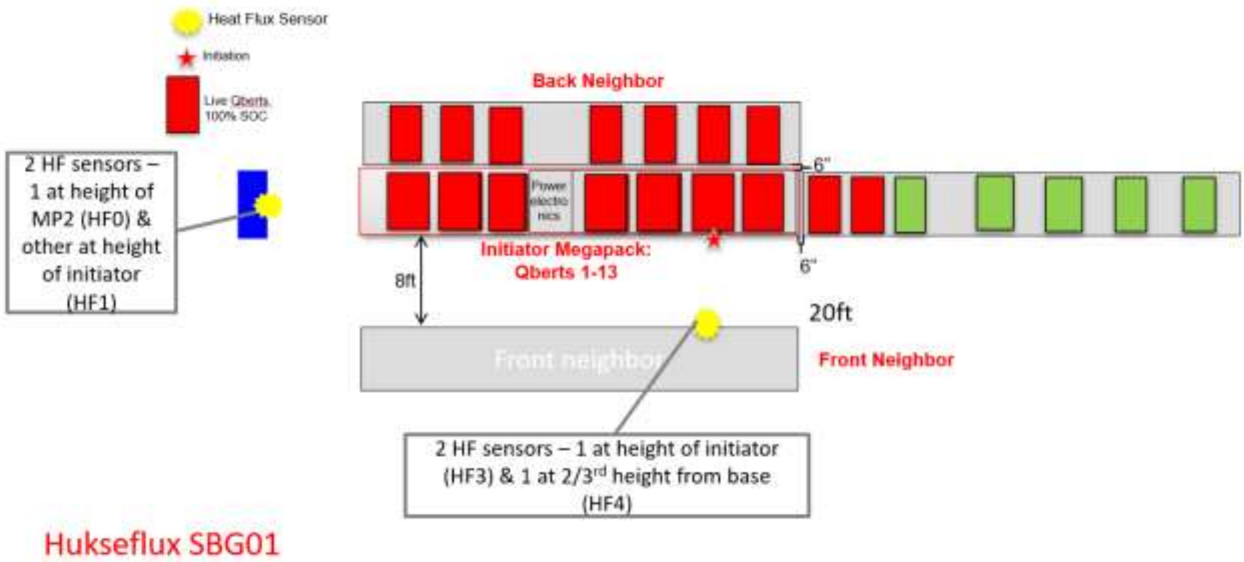

Fig 5. Heat Flux locations

Fig 6. Camera Layout

Fig 7. Pre test photos

Fig 8. Post-test photo of Initiator location

Fig 9. Internal view, post test photo

Fig 10. Initiator Qbert module, Rear view – Post test

Fig 11. Front view – Post test

Fig 12. Side view

Fig 13.– Initiator Battery module – Post test

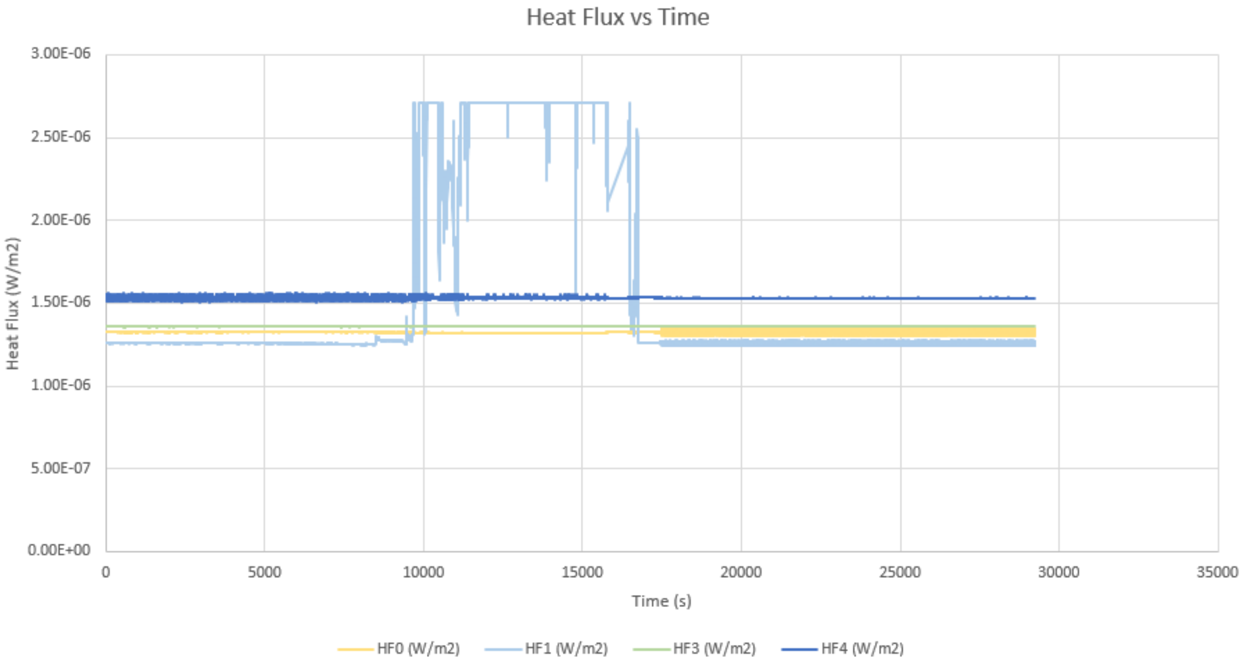

Fig 14.- Heat Flux graph

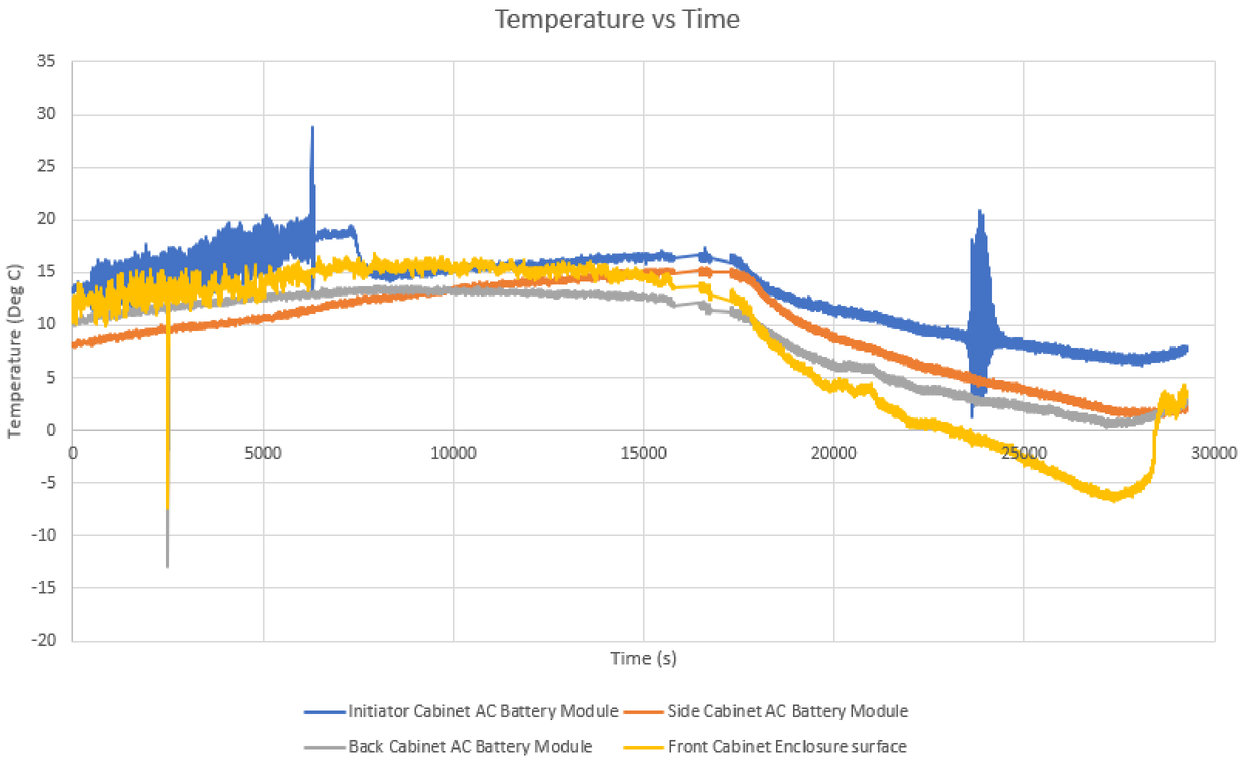

Fig 15.- Temperature graph – Target BESS and Initiator BESS non-initiating AC Battery modules

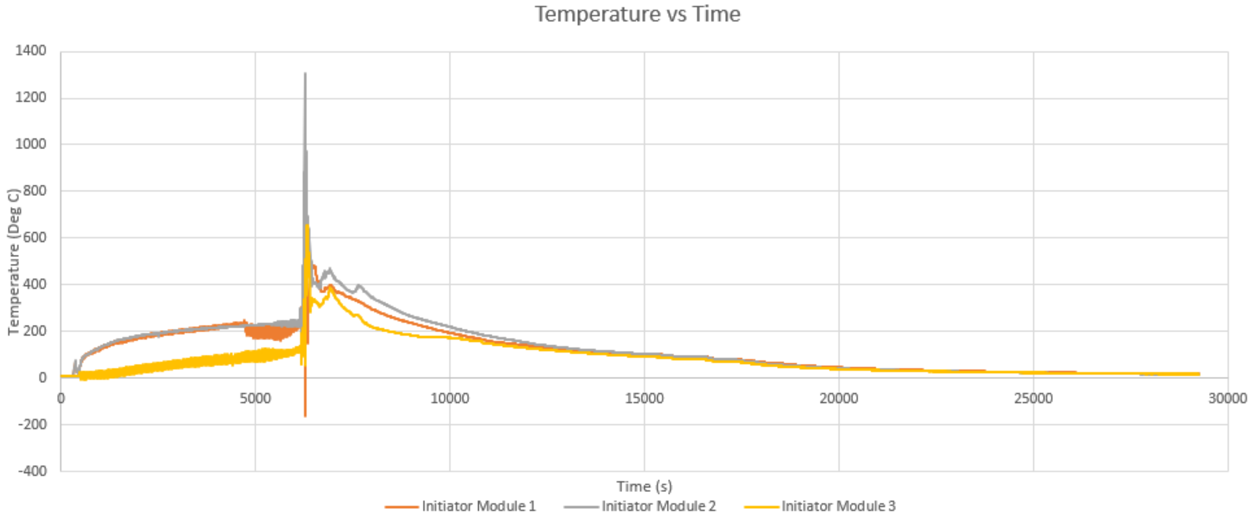

Fig 16.- Temperature graph – Initator module tray

Table 1. Test results per Clause 9.7

#	Items	Description	Comment
a)	Unit manufacturer name and model number (and whether UL 9540 compliant);	Tesla, Megapack 2, 174884-XX-Y, UL9540 compliant	Pass
b)	Number of modules in the initiating BESS unit;	3 module forming AC Battery and 19 AC Batteries in Megapack	Pass
c)	The critical construction of the initiating BESS unit per 5.3;	UL 9540 compliant.	Pass
d)	Fire protection features/detection/suppression systems within unit;	Optional Detection Signal	Pass
e)	Module voltage(s) corresponding to the tested SOC;	100%SOC measured. 373.5V Module voltage	Pass
f)	The thermal runaway initiation method used;	4 film heaters heating 6 cells simultaneously	Pass
g)	Location of the initiating module within the BESS unit;	Initiator Megapack AC Battery 7-3	Pass
h)	Diagram and dimensions of the test setup including mounting location of the initiating and target BESS units, and the locations of walls, ceilings, and soffits;	6 inches from ISO knuckle of Initiating unit to side and back Target unit. 8 ft from front Target.; wall – 5 ft from Initiator	Pass
i)	Observation of any flaming outside the initiating BESS enclosure;	No	Pass
j)	Chemical and convective heat release rate versus time data;	N/A	N/A
k)	Separation distances from the initiating BESS unit to target walls (e.g. distances A and C in Figure 9.1);	5 ft	Pass
l)	Separation distances from the initiating BESS unit to target BESS units (e.g. distances D and H in Figure 9.1);	6 inches from ISO knuckle	Pass
m)	The maximum wall surface and target BESS temperatures achieved during the test and the location of the measuring thermocouple;	Wall – 30.75 deg C Front Target surface-16.8C	Pass
n)	The maximum ceiling or soffit surface temperatures achieved during the indoor or outdoor wall mounted test and the location of the measuring thermocouple;	N/A	N/A
o)	The maximum incident heat flux on target wall surfaces and target BESS units;	Wall-2.71 X 10^-6 W/m2 Front Target – 1.54 X 10^-6 W/m2	Pass
p)	The maximum incident heat flux on target ceiling or soffit surfaces achieved during the indoor or outdoor wall mounted test;	N/A	N/A
q)	Gas generation and composition data;	N/A	N/A
r)	Peak smoke release rate and total smoke release data;	N/A	N/A
s)	Indication of the activation of integral fire protection systems and if activated the time into the test at which activation occurred;	N/A	N/A
t)	Observation of flying debris or explosive discharge of gases;	None observed	Pass
u)	Observation of re-ignition(s) from thermal runaway events;	None observed	Pass
v)	Observation(s) of sparks, electrical arcs, or other electrical events;	None observed	Pass
w)	Observations of the damage to: 1) The initiating BESS unit;	1. No external damage. Initiating AC battery	Pass

	2) Target BESS units; and 3) Adjacent walls	module had burn marks. Other 2 modules of AC Battery intact. Balance 18 AC AC Battery intact. 2. None observed 3. None observed	
x)	Photos and video of the test	Attached	Pass
Note:			

- End of Report -